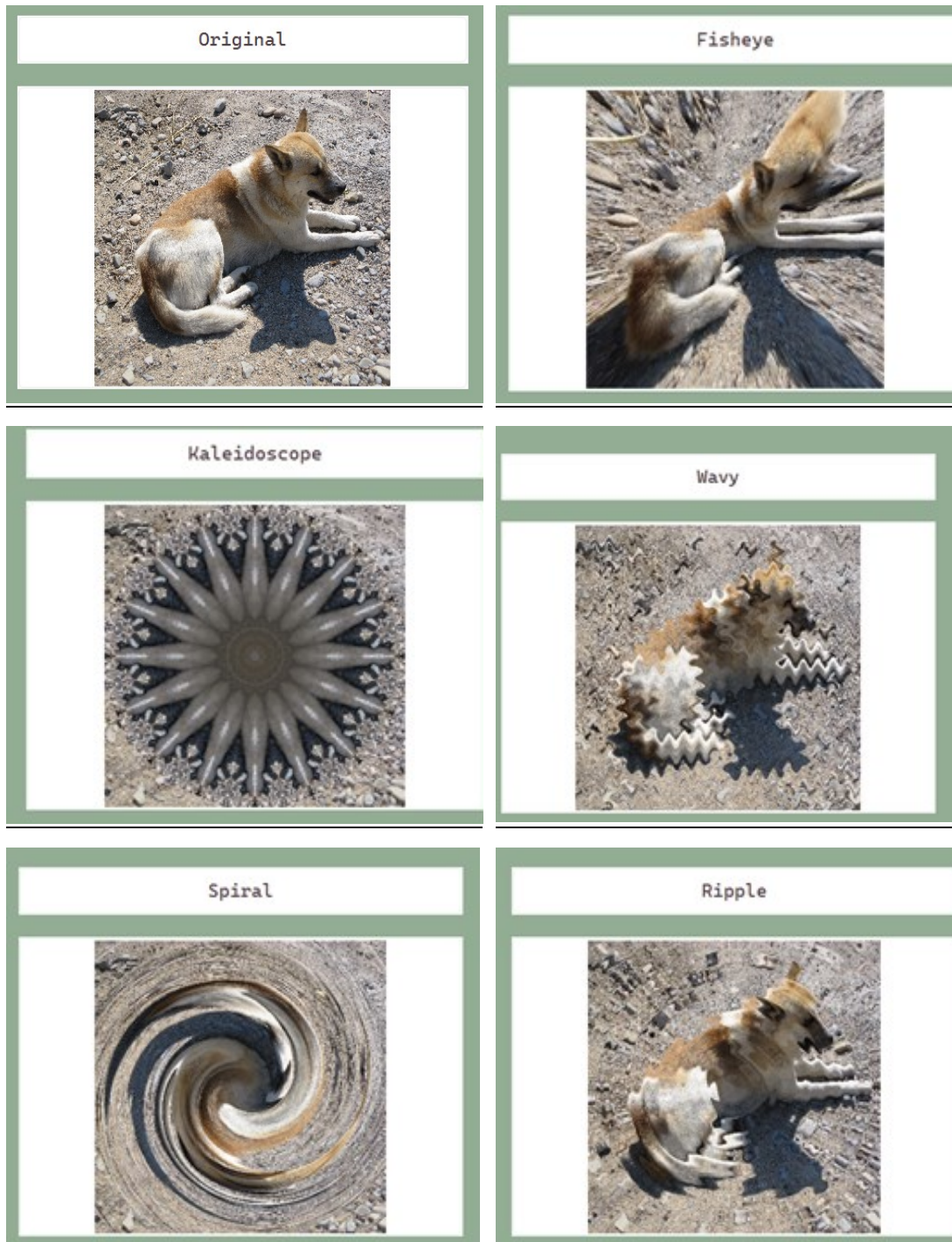


Digital Image Processing

Homework 6

PART1: Geometric Transformation

Design a computer program for geometric transformation of an image.



Fish eye:

1. calculate each pixel's distance and angle from the center using trigonometry
2. scale the distance using transformation formula, making the pixels closer to the center be more distorted (adjust the intensity parameter)
3. reassign the transformed pixels within bounds and display results

Kaleidoscope:

1. calculate for the center coordinates
2. intensity decides image center coordinates and the angular width of each segment is $2 * M_PI / \text{segments}$
3. adjust the pixel's angle to be symmetrical within a segment
4. reassign the transformed pixels within bounds and display results

Wavy:

1. define each pixel's x.y coordinates shifted using sine function(frequency and intensity)
2. calculate the amount of shift by the row(y) and column(x) for every pixel
3. reassign the transformed pixels within bounds and display results

Spiral:

1. calculate each pixel's distance and angle from the center
2. increment the angle by the pixel's distance from the center(distance*intensity)
3. reassign the transformed pixels within bounds and display results

Ripple:

1. calculate each pixel's distance and angle from the center
2. add a sinusoidal ripple to the pixel's distance, and the frequency and amplitude are determined by 0.1 and intensity.
3. reassign the transformed pixels within bounds and display results

PART2: Image Fusion Using Wavelet Transform

The algorithm of image fusion using DWT:

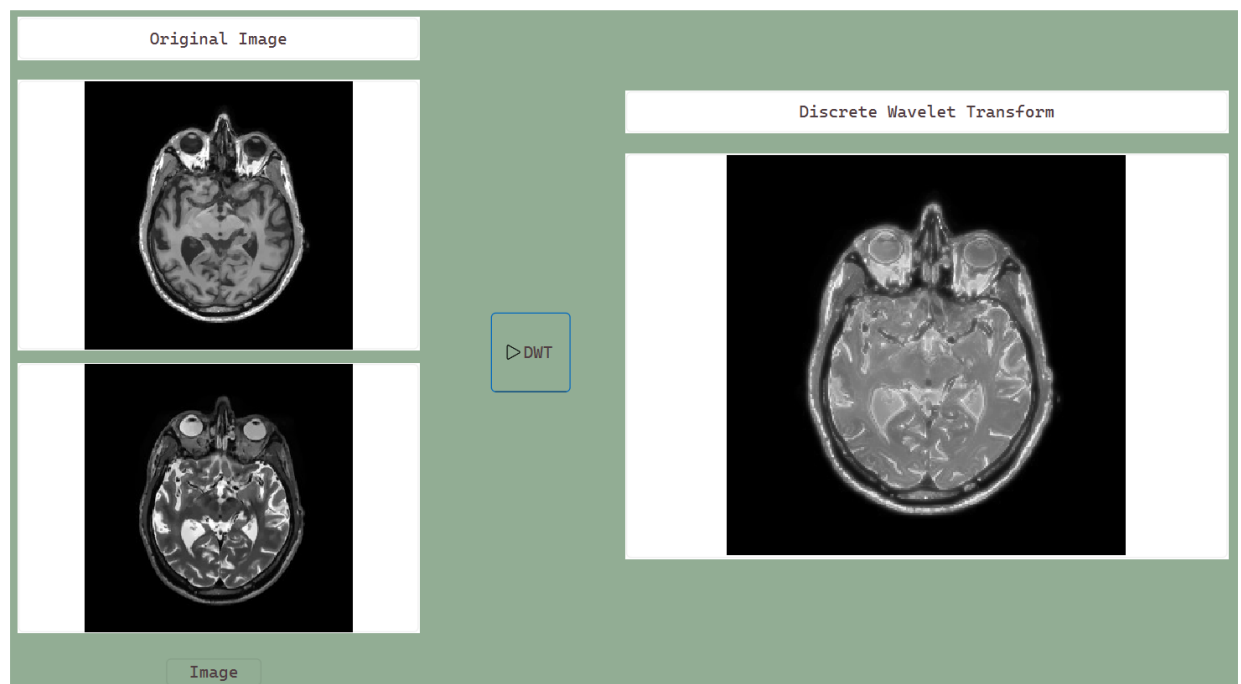
1. The size of images to be fused needs to be the same size and the resolution needs to be of power of two.
2. The two dimensional Discrete Wavelet Transform (DWT) should be applied to the

resized images.

3. Fusion rule: The most used image fusion rule using wavelet transform is maximum selection, by comparing the DWT coefficients of the two (or more) images and select the maximum.

4. After selecting the fused low frequency and high frequency bands, fused image is reconstructed

LL (approximation), LH (horizontal details), HL (vertical details), and HH (diagonal details)

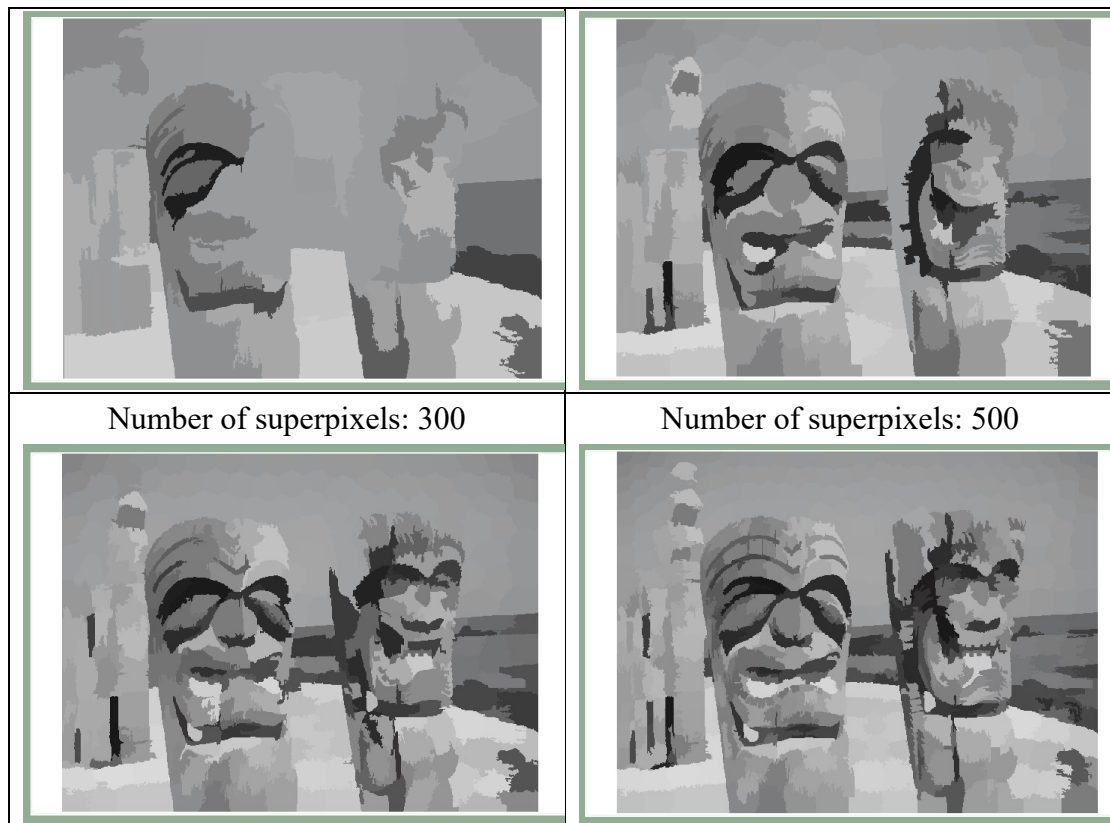


PART3: Superpixel-based Regional Segmentation

1. Choose an image and implement a superpixel algorithm using Simple Linear Iterative Clustering (SLIC)

2. Tune the algorithm's parameters (number of superpixels) for the best results.

Number of superpixels: 150	Number of superpixels: 200
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1. each superpixel is assigned to a cluster center based on color and spatial coordinates.
 - Color distance: Difference in RGB.
 - Spatial distance: Difference in pixel coordinates.
2. Compare neighbor pixels to detect boundaries between clusters
 - Set the boundaries in white(255, 255, 255)